

You are generating a complete browser game for a Mac laptop.

Your objective is to output a single, fully functional, self-contained HTML file containing HTML, CSS, and modern ES6 JavaScript. The file must implement a detailed 3D first-person battlefield training game in which the player conducts electromagnetic-spectrum and sensor missions inside a tense near-future military environment.

The experience should combine the immediacy of a modern tactical FPS with hands-on lessons in:

- detecting electromagnetic emissions
- measuring frequency, bandwidth, power, and timing
- classifying signal types
- associating signals with physical devices and battlefield activity
- using directional measurements to locate emitters
- distinguishing intentional transmissions from unintended electronic noise
- diagnosing communications interference
- understanding what friendly emissions reveal
- distinguishing detection, identification, localization, decoding, authentication, and authorized control

The educational mechanics must be essential to mission success. Do not make the sensor content into optional text, trivia questions, or disconnected quiz screens.

The game should feel like the player is physically carrying instruments, moving through a battlefield, taking measurements, forming hypotheses, and acting on those conclusions.

Do not imitate any specific copyrighted game map, character, logo, weapon design, dialogue, or audiovisual asset. Use an original gritty near-future military aesthetic with dense environmental storytelling, realistic materials, damaged infrastructure, tactical equipment, and atmospheric lighting.

OUTPUT DIRECTIVE

Output only the complete raw HTML document.

Do not include:

- conversational preamble
- markdown fences
- pseudocode
- truncated sections
- TODO comments
- placeholder markup
- explanations outside the HTML
- instructions telling the user to finish code later

The final output must begin with:

```
<!DOCTYPE html>
```

The result must be directly saveable as:

```
battlefield_spectrum_fps.html
```

It should run in a modern Chromium-based browser on macOS.

Use standard CDN-hosted dependencies only. Prefer stable pinned versions from jsDelivr or unpkg.

1. SINGLE-FILE DEPLOYMENT REQUIREMENTS

The entire game must exist in one HTML file.

External dependencies may include only browser-ready CDN scripts such as:

- Three.js
- PointerLockControls
- EffectComposer and standard Three.js post-processing passes
- optional lightweight physics, provided it is loaded from a stable CDN
- optional small browser-compatible utility libraries

Do not use:

- external image files
- external audio files
- GLTF, GLB, OBJ, FBX, or other model files
- external texture files
- remote JSON data
- build tools
- Node.js
- React
- separate CSS or JavaScript files
- APIs requiring credentials

All geometry, textures, sounds, icons, effects, maps, materials, mission data, and UI elements must be generated inside the HTML at runtime.

If browser security prevents reliable operation from a file:// URL, include a clearly visible startup message explaining that the user should run:

```
python3 -m http.server 8000
```

and open:

http://localhost:8000/battlefield_spectrum_fps.html

The game itself must still remain one HTML file.

2. CORE PLAYER EXPERIENCE

The game must begin with a polished title and briefing screen.

Title:

SPECTRUM FRONT

Subtitle:

Anything That Emits Is a Source

The player is a spectrum operations specialist attached to a small tactical unit moving through a damaged urban district.

The player should not begin with every instrument. Equipment is collected from a field equipment case during the opening tutorial.

The primary gameplay loop is:

1. Receive a mission objective by radio.
2. Enter a physical location.
3. Choose an instrument from the equipment wheel.
4. Take measurements from useful positions.
5. Record evidence automatically in a field notebook.
6. Form a conclusion using the evidence.
7. Act on that conclusion.
8. Observe whether the action worked.
9. Receive a short after-action explanation tied to what actually happened.

The player should be able to walk past a signal source without recognizing it, take poor measurements, choose a misleading bearing, misclassify interference, or make an unsupported conclusion.

Wrong decisions must produce believable operational consequences rather than simply displaying "Incorrect."

Examples:

- pursuing the wrong emitter wastes mission time
- a poor bearing sends the squad to the wrong building
- excessive friendly transmitter power reveals the command post
- a frequency reassignment fixes one link but disrupts another
- assuming an emitter is hostile creates an unnecessary confrontation
- attempting to command a transmit-only device produces no response
- misdiagnosing interference leaves medical telemetry unreliable

3. ORIGINAL BATTLEFIELD ENVIRONMENT

Create a dense, original, multi-zone urban battlefield approximately 180 by 180 game units.

The environment should contain interconnected locations rather than one empty arena.

Required zones:

1. Forward Operating Base
2. equipment tent
3. tactical map table
4. radio mast
5. generator
6. command vehicle
7. equipment cases
8. friendly network nodes
9. Damaged Residential Block
10. two-story structures
11. balconies
12. stairways
13. courtyards
14. partially collapsed walls
15. rooftop access
16. reflective surfaces that create multipath
17. Medical Aid Station
18. telemetry equipment

19. field beds
20. emergency lighting
21. intermittent communications problem
22. sensitive receiver affected by nearby emissions
23. Industrial Yard
24. shipping containers
25. generators
26. motors
27. switching power supplies
28. machinery producing both intentional and unintended emissions
29. Communications Ridge
30. relay tower
31. high ground
32. long-distance radio links
33. line-of-sight and obstruction challenges
34. Abandoned Market
35. dense clutter
36. signs
37. stalls
38. narrow passages
39. multiple possible emitter hiding locations
40. Enemy Observation Post
41. hostile sensor equipment
42. autonomous drone pad
43. passive monitoring equipment
44. command antenna
45. guarded access
46. Final Command Sector
47. multiple simultaneous signals

- 48. active interference
- 49. damaged infrastructure
- 50. dynamic mission conclusion

Include:

- streets
- alleys
- sandbags
- barriers
- rubble
- utility poles
- cables
- shattered windows
- stairs
- rooftops
- ladders or ramps
- interior rooms
- cover positions
- smoke
- fires
- moving fans
- flickering lights
- distant tracer-like visual effects
- animated flags, wires, foliage, or fabric

The map must support meaningful position-dependent sensor measurements.

4. GRAPHICS AND LIGHTING

Use a polished original near-future military visual style.

Implement:

- ACESFilmic or Unreal tone mapping
- configurable exposure
- directional sunlight
- soft shadows
- ambient and hemisphere lighting
- localized point lights
- emergency red lighting
- flickering fire or damaged electrical lights
- emissive instrument screens
- fog or atmospheric haze
- film grain
- vignette

- restrained bloom
- subtle chromatic aberration or color grading
- screen shake during nearby impacts
- animated dust and smoke particles

Procedurally generate materials for:

- cracked concrete
- wet pavement
- painted metal
- rusted metal
- tactical polymer
- dirty glass
- fabric
- soil
- rubble
- road markings
- wooden crates
- electronic displays

Generate textures using JavaScript Canvas.

Where practical, generate:

- albedo maps
- roughness maps
- normal-like bump maps
- emissive maps
- dirt masks
- damage masks

Do not rely on remote texture assets.

Use instancing for repeated rubble, grass, debris, cartridges, or structural elements.

5. FIRST-PERSON CONTROLLER

Implement smooth PointerLock first-person controls.

Required movement:

- WASD movement
- mouse look
- jump
- sprint with Shift
- crouch with Control or C

- smooth acceleration and deceleration
- gravity
- collision with architecture
- stairs or ramps
- camera bob
- subtle movement tilt
- crouch-height interpolation
- sprint field-of-view effect
- landing response

Required controls:

- Left click: use active instrument or fire weapon
- Right click: aim down sights or precision instrument mode
- E: interact
- F: switch between weapon and last-used instrument
- Q: open equipment wheel
- R: reload weapon or reset/retune relevant instrument
- Tab: field notebook
- M: tactical map
- H: contextual hint
- Escape: release pointer lock and pause

Display these controls on the initial briefing screen.

6. PLAYER EQUIPMENT

The player carries both a defensive weapon and specialized sensing tools.

Defensive weapon

Create an original compact tactical carbine or SMG using procedural geometry.

Required features:

- first-person viewmodel
- hands or gloves made from primitives
- idle sway
- walking sway
- sprint pose
- recoil
- reload animation
- muzzle flash
- shell ejection
- impact sparks and dust

- persistent bullet marks
- limited ammunition
- ADS interpolation

The shooting system must support the battlefield atmosphere, but most missions must not be solvable by shooting alone.

Friendly and civilian devices must not be valid targets.

Spectrum analyzer

The analyzer should have an in-world handheld screen showing:

- frequency spectrum
- current center frequency
- span
- signal peaks
- approximate power
- noise floor
- adjustable gain
- marker frequency
- bandwidth estimate

Controls should permit:

- tuning frequency
- changing span
- selecting a signal
- saving a measurement
- marking a suspected emitter

The display must respond to:

- player position
- source distance
- obstruction
- antenna orientation
- source state
- active interference
- noise floor

Directional antenna

Create a handheld directional antenna or compact tactical direction finder.

It should show:

- current heading
- relative received power
- bearing-quality meter
- polar response indicator
- strongest observed direction
- multipath warnings when appropriate

The player must physically rotate and move to take bearings.

A single bearing must not reveal the exact source.

Localization should require measurements from separated positions.

Near-field probe

Create a handheld probe for inspecting nearby electronics.

It should react strongly only at close range.

Use it to distinguish:

- intended wireless transmission
- switching power-supply noise
- motor-controller noise
- faulty LED driver
- cable leakage
- digital-clock harmonics

Network diagnostic tablet

The tablet should display a simplified friendly network:

- nodes
- links
- signal quality
- packet loss
- latency
- channel assignments
- acknowledgements
- link failures

The player can use it only on authorized friendly systems.

Field map

The tactical map should support:

- player position
- mission areas
- known friendly systems
- manually placed measurement points
- bearing lines
- probability/search regions
- suspected source markers
- coverage overlays

Do not automatically reveal unknown emitters.

Field notebook

Automatically record measurements with:

- time
- player location
- instrument
- frequency
- power
- bearing
- bandwidth
- observation
- confidence
- associated mission

Allow the player to open the notebook and choose from concise conclusions based on accumulated evidence.

Only conclusions supported by the collected evidence should count as well justified.

7. SIGNAL SIMULATION

Implement a simplified but coherent electromagnetic simulation.

Each emitter must have:

- unique ID
- physical position
- center frequency
- bandwidth

- transmit power
- duty cycle
- transmission pattern
- source type
- antenna pattern
- active state
- harmonics if applicable
- friendly, civilian, unknown, or hostile classification
- intentional or unintended emission type

Required emitters:

1. Friendly command radio
2. Friendly medical telemetry link
3. Friendly drone-control link
4. Friendly relay beacon
5. Civilian environmental sensor
6. Civilian wireless camera
7. Faulty switching power supply
8. LED controller producing harmonics
9. Industrial motor controller
10. Unknown periodic beacon
11. Hostile observation drone
12. Hostile command transmitter
13. Hostile sensor node
14. Decoy transmitter
15. Transmit-only beacon
16. Authenticated friendly remote device

Model received strength using a simplified link equation incorporating:

- source power
- inverse-distance or logarithmic path loss
- obstruction loss
- antenna directionality
- player antenna orientation
- local noise
- limited multipath effects
- optional fading variation

The simulation does not need research-grade fidelity, but it must be internally consistent.

Moving closer should generally increase signal strength.

Walls and terrain should generally reduce signal strength.

Directional measurements should depend strongly on player heading.

Reflective urban geometry should occasionally produce misleading local peaks.

8. EDUCATIONAL MISSION CAMPAIGN

Include a complete playable campaign with at least six missions.

Each mission must take place in the same persistent battlefield.

The player should retain unlocked tools and notebook history.

Mission 1: The Room Is Alive

Location:

Forward Operating Base

Situation:

A new operations center has been assembled. Multiple signals are visible, and one is disrupting staff equipment.

Objectives:

- collect the spectrum analyzer
- scan at least three frequency ranges
- measure four different emissions
- distinguish communications signals from electronic noise
- identify the source correlated with a flickering field light
- inspect it using the near-field probe
- report whether it is a transmitter or faulty electronics

Teaching goals:

- frequency
- bandwidth
- power
- noise floor
- continuous versus burst activity
- intentional versus unintentional emission

The correct source should be a damaged LED power controller producing broadband and harmonic noise.

Mission 2: What Is the Device Doing?

Location:

Industrial Yard

Situation:

A remote pump station is transmitting irregularly. The team needs to know whether it is functioning, reconnecting, or failing.

Objectives:

- observe the pump over time
- identify its normal periodic report
- recognize a command-response sequence
- detect a transition into repeated reconnection attempts
- choose the best-supported device-state interpretation
- use the authorized network tablet to confirm the diagnosis

Teaching goals:

- state inference
- periodicity
- request-response timing
- correlation versus causation
- confidence and alternative hypotheses

Mission 3: Find the Emitter

Location:

Damaged Residential Block and Market

Situation:

An unknown periodic beacon appeared after civilians evacuated the area.

Objectives:

- identify its frequency
- take at least three directional measurements from separated positions
- avoid relying on one reflected bearing
- place bearing lines on the map
- define a search region
- locate and inspect the device

- classify it before deciding what to do

The device should be a forgotten civilian environmental monitor rather than a hostile system.

Teaching goals:

- bearing
- triangulation
- multipath
- localization uncertainty
- unknown does not mean hostile

Mission 4: Interference at the Aid Station

Location:

Medical Aid Station

Situation:

Patient telemetry becomes unreliable whenever the logistics team operates nearby.

Objectives:

- reproduce the outage
- compare the medical signal and nearby logistics transmissions
- measure occupied bandwidth
- determine whether the failure is weak coverage, co-channel interference, adjacent-channel interference, overload, or equipment failure
- test at least two hypotheses
- apply an authorized channel or power adjustment
- verify restored service

Teaching goals:

- interference diagnosis
- controlled testing
- receiver overload
- adjacent-channel effects
- verification after repair

Mission 5: What Are We Revealing?

Location:

Communications Ridge and Forward Operating Base

Situation:

A hostile passive observer appears to have identified the location of the friendly command post.

Objectives:

- enter temporary passive-observer mode
- examine friendly traffic timing and signal strength
- identify the most conspicuous friendly node
- determine what behavior reveals command activity
- return to the friendly network
- reduce unnecessary beaconing or excessive power
- adjust relay topology or antenna direction
- preserve required service while reducing observability

Teaching goals:

- electromagnetic signature
- metadata
- network centrality
- power discipline
- functionality versus observability

Mission 6: Detection Is Not Control

Location:

Enemy Observation Post and nearby utility compound

Situation:

Four emitting systems have been detected. Leadership asks which can be controlled.

Devices:

- transmit-only sensor beacon
- authenticated friendly pump controller
- authorized course-owned rover
- noisy power supply

Objectives:

- detect all four
- inspect each device
- determine which devices have receivers
- distinguish addressing from authentication
- successfully command only the authorized rover

- diagnose one induced control failure
- reject unsupported claims that every emitter can be controlled

Teaching goals:

- detect
- identify
- locate
- decode
- address
- authenticate
- authorize
- control
- acknowledge

Final Mission: Spectrum Front

Location:

Entire battlefield

Situation:

A coordinated assault and communications crisis occur simultaneously.

Dynamic events:

- medical telemetry failure
- hostile observation drone launch
- unknown new emission
- friendly relay failure
- increased drone bandwidth demand
- misleading report of “jamming”
- remote gate or warning device failing to acknowledge commands
- hostile command transmitter moving between locations

The player must:

- triage incidents
- use the correct instrument for each problem
- preserve critical friendly services
- locate at least one moving emitter
- distinguish actual interference from ordinary equipment failure
- reduce friendly signature
- use the defensive weapon against armed hostile combatants or drones where necessary
- avoid shooting civilian or friendly equipment
- provide a final evidence-based mission assessment

The ending should depend on:

- quality of measurements
 - number of supported conclusions
 - time used
 - medical network uptime
 - friendly signature exposure
 - civilian-device handling
 - successful localization
 - unnecessary use of force
 - final briefing accuracy
-

9. FIELD LEARNING SYSTEM

Do not use traditional multiple-choice quiz screens.

Learning should occur through:

- instrument feedback
- radio dialogue
- physical measurement
- consequence
- short contextual explanations
- after-action replay

When the player first observes an important phenomenon, briefly display a compact lesson card.

Examples:

After measuring bandwidth:

"Bandwidth is the range of frequencies occupied by a signal. A strong center peak does not describe the entire channel it consumes."

After receiving conflicting bearings:

"Urban reflections can produce a stronger local direction than the direct path. Change position before trusting the bearing."

After detecting a signal from a transmit-only beacon:

"Detection establishes that energy is present. It does not establish that the device has a receiver or accepts commands."

Keep lesson cards short and directly tied to the player's measurement.

Include an optional H key hint system with three levels:

1. strategic hint
2. instrument suggestion
3. explicit procedural guidance

Do not reveal the answer immediately.

10. NPCs AND COMBAT

Include simple but functional NPC behavior.

Friendly NPCs:

- remain near assigned posts
- provide contextual mission dialogue
- indicate operational consequences
- do not block key routes

Hostile NPCs:

- patrol
- investigate sound
- take cover
- fire at the player
- can be engaged with the weapon
- should not appear in every educational measurement sequence

Hostile drones:

- patrol or launch when triggered
- emit detectable command or telemetry signals
- can be located electromagnetically
- may be disabled through combat after identification

Civilian or neutral equipment must be visually and logically distinct from valid hostile targets.

The player should be penalized for unnecessary destruction of civilian or friendly devices.

11. PHYSICS AND DESTRUCTION

Implement lightweight physics sufficient for:

- player collision
- gravity
- jumping
- pushable crates or barrels
- projectile raycasts
- debris impulses
- destructible small props
- glass fragments
- sparks
- smoke
- persistent impact marks

Do not attempt an overly heavy universal physics simulation if it harms performance.

Prioritize reliable gameplay and collision over maximum physics complexity.

12. SOUND

Because external audio files are prohibited, generate sound procedurally using the Web Audio API.

Required sounds:

- weapon fire
- reload clicks
- footsteps by surface type
- equipment pickup
- spectrum detector chirps
- signal-strength tone
- radio static
- UI confirmation
- nearby electrical hum
- generator rumble
- drone motor
- distant impacts
- alarm
- mission-complete tone

Include volume controls.

Do not play sound until after a user gesture.

13. USER INTERFACE

Required HUD elements:

- health
- ammunition
- active equipment
- current mission objective
- contextual interaction prompt
- signal-strength indicator when using instruments
- mission timer when relevant
- compact compass
- notification feed

Instrument screens should appear physically attached to the first-person equipment view when practical.

Required menus:

- title and briefing
- pause
- controls
- graphics settings
- audio settings
- mission notebook
- tactical map
- equipment wheel
- after-action report

Graphics settings:

- low
- medium
- high

Settings should adjust:

- shadow resolution
- particle count
- post-processing
- pixel ratio
- draw distance
- bloom
- environmental detail

Default to settings suitable for a modern Mac laptop rather than maximum GPU load.

14. PERFORMANCE REQUIREMENTS

Target smooth play on a modern Mac laptop.

Optimize for approximately 60 FPS on medium settings.

Required optimizations:

- instanced meshes for repeated geometry
- limited shadow-casting lights
- restrained shadow-map resolution
- pooled particle systems
- pooled bullet decals
- frustum culling
- simplified distant geometry
- capped device pixel ratio
- reusable materials and geometries
- no unbounded object creation in the animation loop
- no large per-frame memory allocations
- dynamic quality settings

Keep all important gameplay functional when expensive post-processing is disabled.

15. ROBUSTNESS REQUIREMENTS

The final HTML must:

- load without JavaScript syntax errors
- display a useful loading screen
- report dependency failures visibly
- prevent the UI from permanently blocking pointer lock
- allow the player to close every modal
- handle pointer-lock loss gracefully
- support restarting the current mission
- support resetting the entire campaign
- avoid null-reference errors when optional effects are disabled
- avoid duplicate animation loops
- resize correctly when the browser window changes
- prevent default browser context menus during right-click ADS
- prevent accidental page scrolling during gameplay
- store basic progress and settings in localStorage
- include a “Reset saved progress” control

The start button must visibly work.

After clicking Start:

- the briefing overlay must close
- the 3D scene must become interactive
- clicking the scene must activate pointer lock
- the player must be able to move
- the opening equipment case must be reachable
- the first mission objective must appear

16. CODE ORGANIZATION

Although everything must be inside one HTML file, organize the JavaScript into clear classes or modules.

Suggested structure:

- Game
- RendererManager
- WorldBuilder
- MaterialFactory
- TextureFactory
- AudioManager
- InputManager
- PlayerController
- WeaponSystem
- InstrumentManager
- SpectrumAnalyzer
- DirectionFinder
- NearFieldProbe
- NetworkTablet
- SignalEngine
- PropagationEngine
- NetworkEngine
- MissionManager
- NPCManager
- ParticleManager
- UIManager
- NotebookManager
- SaveManager

Use comments to identify major systems, but do not leave incomplete sections.

17. FINAL QUALITY CHECK

Before producing the HTML, internally verify that:

- all opening and closing HTML tags match
- all imported scripts use stable valid URLs
- all referenced JavaScript classes exist
- Start closes the overlay
- pointer lock can activate
- movement works
- collision works
- at least six missions are playable
- equipment can be equipped
- signal measurements change with position
- directional readings change with player heading
- notebook entries are created
- mission progression can complete
- the final mission can reach an ending
- pause and restart work
- the game remains usable on medium graphics settings

Do not describe this verification process in the output.

Output only the final complete HTML document.